Project Methodology and AI Usage

Corporate Credit Risk Analysis & AI-Assisted Decision Support

Project title: Corporate Credit Risk Analysis & AI-Assisted Decision Support

Company analysed: Mahindra & Mahindra Limited

Classification: Independent Educational Portfolio Project

Analysis period: FY2024–FY2026, with selected FY2027 outlook information from CRISIL's 18 September 2026 rating rationale

Document: 04 of the project portfolio (methodology, data governance and responsible AI use)

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> **Important notice**
>
> This is an independent educational portfolio project built from public information. It is not an official assessment, lending decision or credit rating by any financial institution, and it does not represent the views of Mahindra & Mahindra or CRISIL. See Section 7 for the full disclosure.

1. Project Purpose and Scope

This document explains how the Mahindra & Mahindra Limited credit-risk project was built, how the data were handled, what quality checks apply, and how AI was and was not used. The project is intended to demonstrate financial analysis, credit-risk identification, analytical judgement, documentation discipline, quality assurance and responsible AI-assisted decision support, in support of a Program Analyst application in a financial-services setting.

The analysis covers FY2024–FY2026 and the following areas:

- Revenue and profitability trends; borrowings and liquidity indicators; cash flow and investment outflows.
- Segment performance and Financial Services loan quality.
- Debt maturity and refinancing considerations; external credit ratings.
- Audit key matters and internal-control considerations; credit risks, mitigants and monitoring indicators.

Financial inputs are primarily consolidated annual-report figures. CRISIL's selected financial indicators are standalone and are treated separately.

2. Source and Data-Management Approach

Stage 1 — Source collection

Four public sources were identified: the FY2025–26 annual report, the FY2024–25 annual report, CRISIL's rating rationale dated 18 September 2026, and the FY2025–26 results announcement published by Mahindra & Mahindra Financial Services Limited, the Financial Services subsidiary (full links in Section 9). Every data point should be attributed to its source document and reporting scope (consolidated or standalone) at the point of use. The Mahindra Finance announcement is used only for the Financial Services figures it supports (such as asset quality); it is not a source for group-level consolidated figures, which come from the annual reports.

Stage 2 — Data structuring

Reported data are organised into consistent financial-year columns and metric categories. Each item is tagged to one of five classes so that a reader can tell fact from judgement:

- **Reported figures:** values taken directly from a source document.
- **Calculated ratios:** values derived by formula from reported figures (for example, Stage 3 ratios from disclosed loan balances).
- **External agency opinions:** CRISIL ratings and outlooks, which remain CRISIL's views.
- **Analyst interpretation:** judgements drawn by the project author.
- **Unverified assumptions or open issues:** items awaiting confirmation, logged rather than silently resolved.

3. Financial-Analysis Methodology

Stage 3 — Financial calculations

Consolidated annual-report figures are used for group-level metrics (revenue, profitability, borrowings and cash flow), with the same denominator and period basis in each year. Stage 3 ratios use the separately identified Financial Services loan data. Standalone CRISIL indicators are never merged into these series. Borrowings / assets is defined but not calculated, as total assets were not supplied.

Metric	Formula	Denominator / scope note
Revenue growth	(Revenue FY[t] ÷ Revenue FY[t−1]) − 1	Consolidated revenue from operations
PBT margin	Profit before tax ÷ Revenue from operations	Consolidated; same-year revenue
Consolidated profit margin	Consolidated profit ÷ Revenue from operations	Consolidated; same-year revenue
Borrowings growth	(Borrowings FY[t] ÷ Borrowings FY[t−1]) − 1	Borrowings excluding leases
Borrowings / assets	Borrowings excluding leases ÷ Total assets	Total assets to be taken from the consolidated balance sheet
CFO / revenue	Operating cash flow ÷ Revenue from operations	Consolidated; same-year revenue
Gross Stage 3 ratio	Gross Stage 3 loans ÷ Gross loans	Disclosed Financial Services loans
Net Stage 3 ratio	Net Stage 3 loans ÷ Net loans	Disclosed Financial Services loans

Illustrative recalculation from the core figures supplied for this project (₹ crore inputs; ratios rounded to two decimals):

Metric	FY2024	FY2025	FY2026
Revenue from operations (₹ cr)	139,078.27	159,210.82	197,792.78
Revenue growth	n/a	14.48%	24.23%
PBT margin	11.49%	11.98%	12.73%
Consolidated profit margin	8.82%	8.84%	9.41%
Borrowings excl. leases (₹ cr)	103,203.23	121,172.95	129,614.50
Borrowings growth	n/a	17.41%	6.97%
CFO / revenue	−4.05%	−0.18%	5.89%
Gross Stage 3 (as supplied)	~3.87%	~4.07%	~3.42%
Net Stage 3 (as supplied)	~1.68%	~1.87%	~1.44%

Rows other than the two Stage 3 rows were recalculated by arithmetic from the supplied inputs. Stage 3 ratios are reproduced as supplied; the underlying loan balances were not part of the inputs used to prepare this document, so they were not recomputed here.

4. Credit-Risk Assessment Methodology

Stage 4 — Credit interpretation

Profitability, leverage, liquidity, cash flow, debt maturity, segment diversification and Financial Services asset quality are read together rather than in isolation. No single ratio determines creditworthiness. For example, strong profit growth is weighed against rising borrowings, and a sharp cash-flow improvement is weighed against the question of whether it is sustainable.

Stage 5 — Risk and monitoring framework

Each identified risk is linked through a four-part chain so that conclusions can be traced back to evidence:

Element	What it records
Evidence	The specific reported figure, disclosure or agency statement, with source and reporting scope.
Potential credit impact	How the risk could weaken repayment capacity, liquidity or asset quality.
Mitigants	Factors that may offset the risk, such as diversification, ratings support or reported coverage.
Monitoring indicators	Observable measures to track over time, with a stated trigger for re-review.

5. Quality Assurance and Open Issues

Stage 6 — Quality assurance

The table separates checks that have been performed from checks that remain planned. A check is marked complete only where it was actually carried out.

Check	Scope	Status
Calculation accuracy	Revenue growth, margins, borrowings growth and CFO / revenue recomputed by script from supplied inputs	Completed for this document
Arithmetic of flagged gaps	₹845.77 crore and ₹85.04 crore differences recomputed from supplied totals	Completed
Source-to-figure verification	Each supplied figure traced back to the annual reports, CRISIL rationale and results release	Planned; not performed by AI
Stage 3 ratio recalculation	Recompute from disclosed loan balances	Planned
Formula and period alignment	Review of workbook formulas, year labels and denominators	Planned
Cross-deliverable consistency	Same figures across workbook, memo and presentation	Planned
Reporting-scope check	Consolidated and standalone data kept separate	Applied by design; final cross-deliverable verification pending

Open issues and limitations

- **Reporting scope.** Annual-report figures are primarily consolidated; CRISIL's selected indicators are standalone. They must not be merged into one ratio series.
- **Revenue definition.** FY2026 consolidated revenue from operations is ₹197,792.78 crore, while the supplied segment external-revenue total is ₹198,638.55 crore (a ₹845.77 crore difference). The two use different source definitions. The discrepancy is preserved and flagged, not reconciled.
- **Debt maturity.** FY2026 maturity-bucket borrowing principal totals ₹129,699.54 crore against balance-sheet borrowings of ₹129,614.50 crore. The ₹85.04 crore difference is an open reconciliation item.
- **Liquidity.** Other bank balances may include restricted, earmarked or margin balances and cannot automatically be treated as unrestricted cash.
- **Cash flow.** The sharp FY2026 CFO improvement is positive, but its drivers and sustainability require further assessment.
- **Financial Services.** Improving Stage 3 ratios do not remove risk. Loan growth, ECL coverage, underwriting, collections and loss severity should be assessed together.
- **Credit cost.** Disclosed impairment balances alone are not sufficient to calculate a valid annual credit-cost ratio, so none is presented.

6. Responsible AI and Ethical Considerations

Where AI can help

AI assistance is suited to supporting tasks: organising supplied data, drafting and editing text, structuring the analysis, suggesting ratio checks, flagging possible inconsistencies, drafting a monitoring framework, and improving the clarity of written and visual communication.

Limits and controls

- AI can misread source tables or make calculation errors, so outputs need analyst review and re-calculation.
- AI-generated conclusions are drafts for the analyst to accept, change or reject.
- AI must not invent missing financial data. Gaps are logged as open issues.
- Source references and reporting definitions must be checked by a person against the original documents.
- Sensitive or confidential information must not be uploaded to an AI tool without authorisation.
- Actual financial decisions require human judgement and professional accountability.

AI did not independently verify the source documents or the source figures used in this project. Verification against the original documents remains a human responsibility, recorded as planned in Section 5.

7. Professional and Ethical Disclosure

- This is an independent educational portfolio project using public information.
- No confidential company data was intentionally used.
- No actual borrower or lending decision was evaluated.
- The project does not represent the views of Mahindra & Mahindra or CRISIL.
- CRISIL ratings are external agency opinions.
- AI assistance does not replace human judgement, source verification or professional accountability.
- The project does not imply employment at, or endorsement by, any named organisation.

8. Limitations and Conclusion

The analysis relies on public disclosures and a limited set of supplied figures. It does not use internal borrower data, covenant terms, collateral information or management access, all of which a real credit decision would require. Several reconciliation items remain open, and source-to-figure verification is still to be completed by the analyst.

Within those limits, the methodology shows a traceable process: attributed sources, separated data classes, consistent formulas, a linked risk-and-monitoring framework, transparent handling of discrepancies, and a clear boundary between AI assistance and human accountability.

9. References

[1] Mahindra & Mahindra Limited. Annual Report FY2025–26. https://www.mahindra.com/annual-report-FY2026/index.html

[2] Mahindra & Mahindra Limited. Annual Report FY2024–25. https://www.mahindra.com/annual-report-FY2025/toc/

[3] CRISIL Ratings. Rating rationale, Mahindra and Mahindra Limited, 18 September 2026. https://www.crisilratings.com/mnt/winshare/Ratings/RatingList/RatingDocs/MahindraandMahindraLimited_September%2018_%202026_RR_404981.html

[4] Mahindra & Mahindra Financial Services Limited (Mahindra Finance). Financial Results – Quarter 4 and Full year FY26, Standalone & Consolidated Results (press release, 24 April 2026). https://www.mahindra.com/news-room/press-release/en/financial-results-quarter-4-and-full-year-fy26-standalone-and-consolidated-results

Links are listed as supplied for the project; tracking parameters have been removed. The link in [4] was opened and confirmed to be the Mahindra Finance announcement, which covers the Financial Services subsidiary rather than group-level results, and is cited only for Financial Services figures; the other links have not been opened. Source figures remain to be verified by the analyst.